CSCI 332, Fall 2026

Homework 10

Due Monday, November 9 Anywhere on Earth (6am Tuesday)

Submission Requirements

- Type or clearly hand-write your solutions into a PDF format so that they are legible and professional. Submit your PDF on Gradescope.
- Do not submit your first draft. Type or clearly re-write your solutions for your final submission. If your submission is not legible, we will ask you to resubmit.
- Use Gradescope to assign problems to the correct page(s) in your solution. If you do not do this correctly, we will ask you to resubmit.

Academic Integrity

Remember, you may access ***any*** resource in preparing your solution to the homework. However, you ***must***

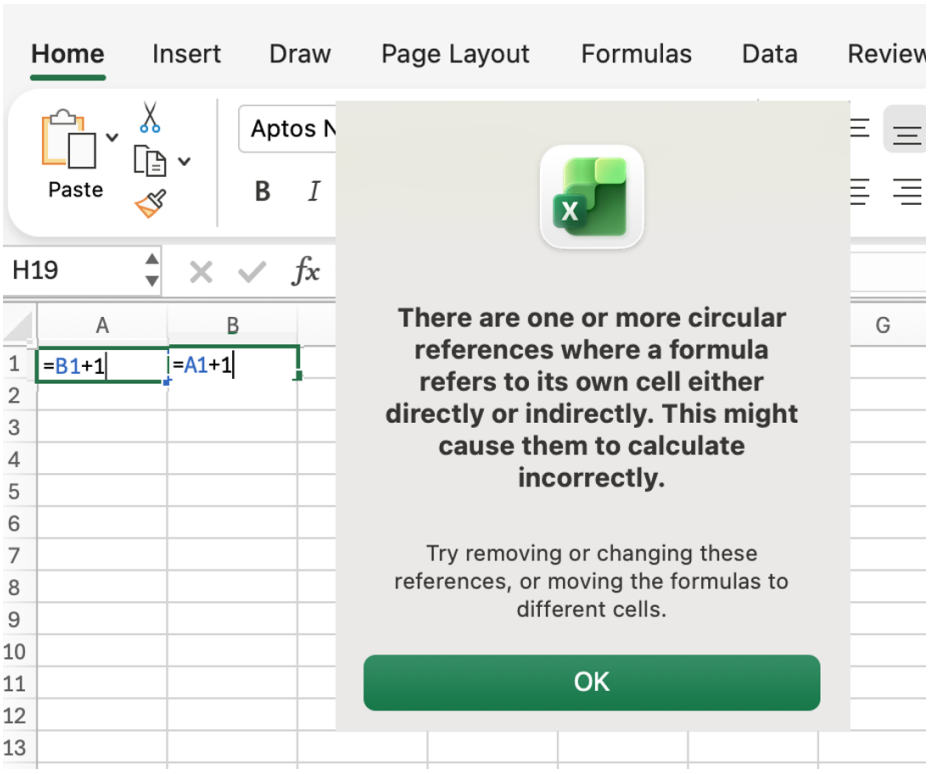
- write your solutions in your own words, and
- credit every resource you use (for example: "Bob Smith helped me on this problem. He took this course at UM in Fall 2020"; "I found a solution to a problem similar to this one in the lecture notes for a different course, found at this link: www.profzeno.com/agreatclass/lecture10"; "Here is a link to the full transcript of my chat with ChatGPT about the problems on this HW".) Ask if you need any clarifications about how to properly cite your sources!

1. (0 points) List the resources you used for each problem in this homework. If you used no resources except lecture notes and videos and the linked textbook, you can say "none". If you used an AI tool, you should put a link to your full conversation here, and you must use the provided prompt to guide the AI to act as a tutor, not an answer-provider.

   You will be asked to resubmit your homework if this section is blank.

   Remember that no matter what resources you consulted, you must write your solutions *in your own words*.

2. In Excel (and other spreadsheet programs), the values of cells can be calculated as functions of the values of other cells. For example, cell $A1$ can be assigned to have value $B1 + 1$. However, if cells reference one another in a circular manner, then the values are undefined, and the program must decide how to handle the error. In Excel, such circular references produce an error:



   You are given the current state of a spreadsheet program as a collection of m strings representing the formulas in each cell. Describe and analyze the runtime of an efficient algorithm to detect whether circular references exist by reducing to a problem on a graph. You may write your answer in paragraph form, as pseudocode, or however you prefer, but you will receive points for clearly addressing each of the following:

   - What are the vertices? What does each vertex represent?
   - What are the edges? Are they directed or undirected?
   - If the vertices and/or edges have associated values, what are they?
   - **What is the precise relationship between your graph and the stated problem?**

- What standard graph algorithm are you using to solve the problem?
- What is the running time of your entire algorithm, *including* the time to build the graph, *as a function of the original input parameters*? (Notice that V and E are *not* parameters of the original problem.)